Supplemental Material

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This document is the appendix to the main paper. Section, equation, and table numbers cited from the main paper follow that manuscript.

Appendix A. Additional Details and Analyses

Appendix A.1. Details of Open-World Datasets

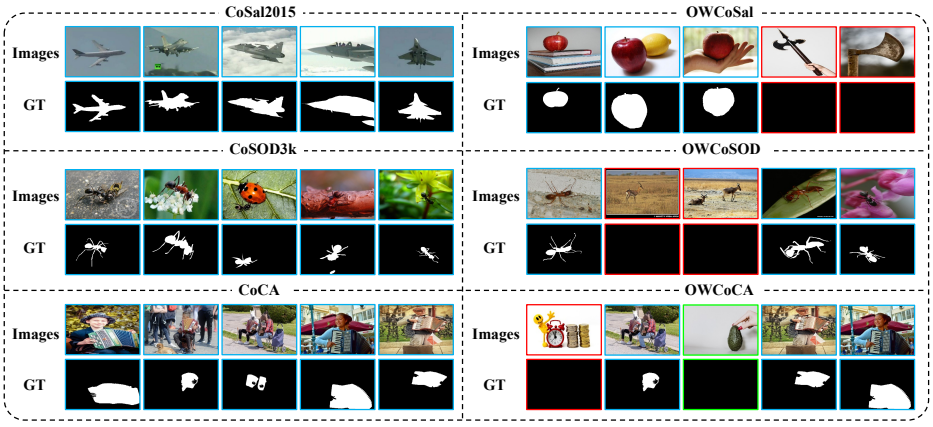


Figure A.1: Samples from six datasets. CoSal2015 [1], CoSOD3k [2], and CoCA [3] all follow the group consensus assumption, while our OWCoSal, OWCoSOD, and OWCoCA contain different types of noise.

The left half of Figure A.1 shows samples from existing datasets. One of the most commonly employed benchmark datasets is CoSal2015 [1], consisting of 2,015 images distributed across 50 categories. Its difficulty mainly stems from images within certain categories that have different backgrounds. CoSOD3k [2] is currently the most extensive evaluation benchmark, featuring a total of 160 categories and more than 3,000 images. In contrast to CoSal2015, CoSOD3k includes a significant number of images with two or three instances to be segmented, covering a wide range of scenes, size variations, optical conditions,

and backgrounds. CoCA [3] represents the most demanding evaluation benchmark, comprising 80 categories with a total of 1,295 images. This dataset includes many highly challenging samples, featuring objects from unknown categories, ghosted objects, extremely noisy backgrounds, and other challenging aspects.

Existing datasets are constructed based on the group consensus assumption, leading to benchmark test sets structured by categories, with each category group containing images featuring the same salient objects, *i.e.*, co-salient objects. Their difficulty is primarily designed from the perspective of individual images, such as multiple scales, complex backgrounds, and interfering objects. Although these designs partially resemble open-world situations, the group consensus assumption itself is overlooked. We argue that it is necessary to break this assumption. In open-world scenarios, the main challenge arises from irrelevant images that do not contain co-salient objects. In application, collected sets of images will inevitably include noisy images, and the objective of CoSOD is to detect co-salient objects, even in the presence of a significant number of noisy images.

To this end, we assemble three new datasets from CoSal2015 [1], CoSOD3k [2], and CoCA [3] for open-world evaluation, which we refer to as OWCoSal, OWCoSOD, and OWCoCA. Specifically, within each category of each dataset, we introduce noisy images in random proportions. We design two types of noise: in the first type, all noisy images added to a group come from a single other category; in the second type, they come from multiple different categories. This approach allows us to more comprehensively simulate real-world scenarios. Reorganizing existing datasets rather than recollecting images offers the following advantages in addition to resource efficiency: **(1) Strict semantic exclusion:** Leveraging rigorous annotations guarantees zero semantic overlap, avoiding the accidental co-saliency that random web images might introduce. **(2) Consistency:** It cleanly isolates "semantic noise" without introducing unwanted low-level domain shifts (*e.g.*, lighting or resolution differences), ensuring that the noise truly originates from different categories. **(3) Real-world dynamics:** We simulate open-world complexity by dynamically varying noise proportions (in OWCoSal and OWCoSOD) and mixing 1 to 3 distinct noise categories per group (in OWCoCA). Consequently, because the injected noise domains are strictly non-overlapping, the model cannot exploit superficial dataset bias. The right half of Figure A.1 shows some examples of OWCoSal, OWCoSOD, and OWCoCA, where borders of different colors indicate different categories.

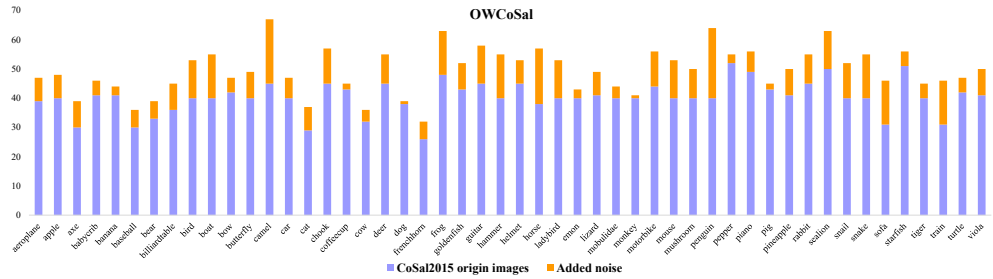


Figure A.2: The structure of OWCoSal.

OWCoSal.. As shown in Figure A.2, building upon CoSal2015 [1], we introduce an additional 460 noisy images from other categories, expanding each category by 2.4% to 37.5%, with the majority concentrated around 18%. In OWCoSal, all noisy images added to a given category originate from a single other category. To control the difficulty of OWCoSal, we keep the number of noisy images at a relatively balanced level, around 18% of the original image count. Meanwhile, we generate the corresponding labels for the noisy images, resulting in 460 all-zero maps.

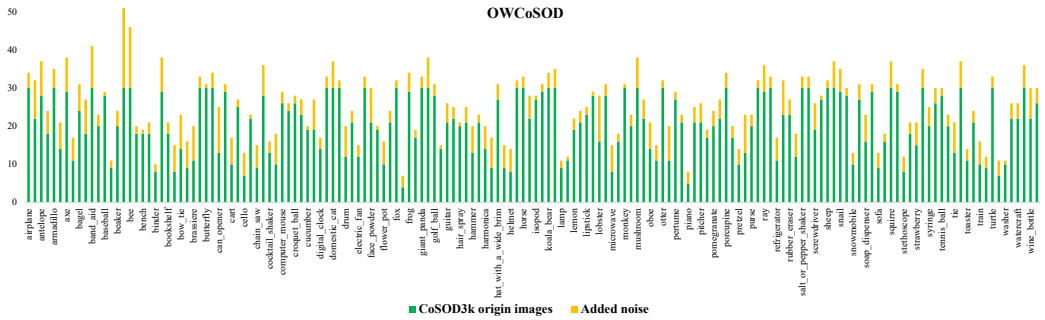


Figure A.3: The structure of OWCoSOD.

OWCoSOD.. We add 739 noisy images to CoSOD3k [2], following the same approach as in OWCoSal, where the noisy images added to each category also originate from a single other category. The expansion ranges from 3.2% to 47.1% for different categories. To distinguish it from OWCoSal, the expansion ratios of most categories in OWCoSOD are concentrated at both ends, around 5% and 40%. The structure of OWCoSOD is shown in Figure A.3.

OWCoCA.. As shown in Figure A.4, OWCoCA extends CoCA [3] by adding 364 noisy images. Unlike OWCoSal and OWCoSOD, the noisy images within

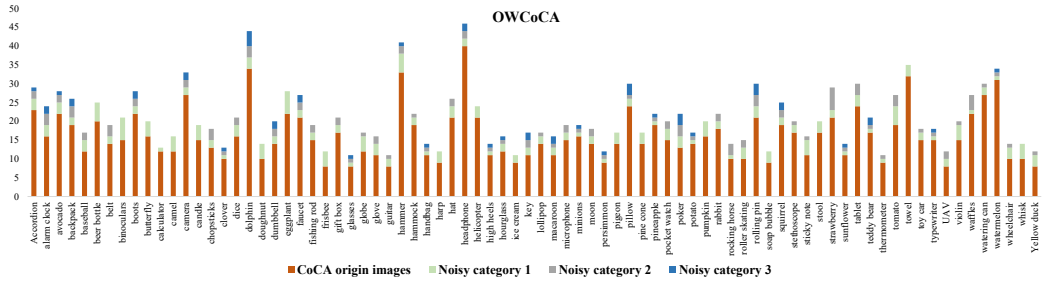


Figure A.4: The structure of OWCoCA.

each group of OWCoCA may originate from different categories. We introduce noise from 1–3 categories per group to simulate open-world scenarios, making OWCoCA the most diverse dataset; combined with the inherent difficulty of CoCA, OWCoCA becomes exceedingly challenging. The corresponding labels are generated in the same way as for OWCoSal and OWCoSOD.

Appendix A.2. Additional Open-World Analysis

To further analyze the model in open-world scenarios, we provide additional visualizations of GSEM and LVGB on noisy image groups.

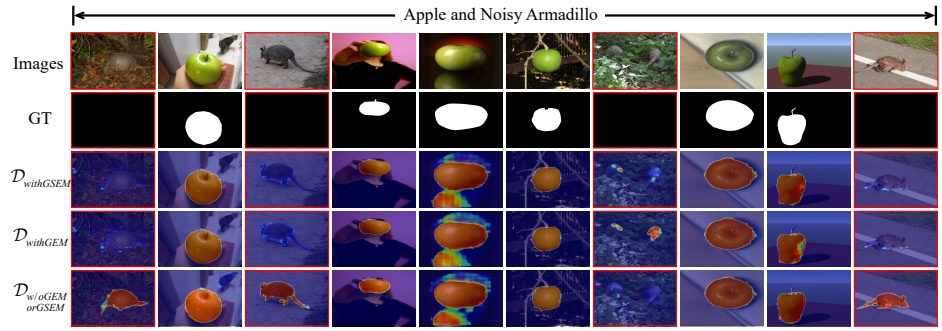


Figure A.5: Decoder response maps obtained with GSEM, with GEM, and with neither of them. The first row shows an image group with noisy images that come from another category. The second row is the ground truth. The third and fourth rows show the response maps $\mathcal{D}_{withGEM}$ and $\mathcal{D}_{w/oGEM}$ from the decoder.

Visual effect of GSEM.. We compare feature visualization results on noisy groups. As shown in Figure A.5, we provide visual results of CoGSEM when "Armadillo" noise is introduced into the "Apple" group. It can be observed that GSEM effectively enhances the model's robustness and outperforms

GEM, allowing the model to better suppress noisy images. However, when GSEM or GEM is not used, the model's robustness significantly decreases, leading to incorrect segmentation of non-co-salient objects in noisy images. This is detrimental to the model's applicability in open-world scenarios.

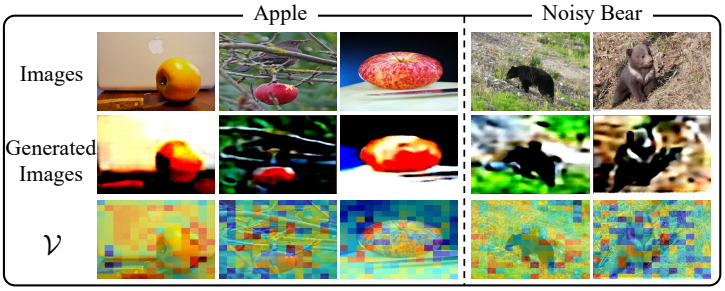


Figure A.6: Image generation results and intermediate feature visualization of the VQ-VAE in the LVGB.

Visual effect of LVGB.. Figure A.6 displays the image generation results in LVGB along with their corresponding feature heat maps in the decoder. As can be observed, LVGB implicitly models group-level global features while accommodating the generation of stochastic features $\mathcal{V}$. For the primary target "Apple", its heat map focuses more on the target region. For the noise "Bear", its heat map is more dispersed, resulting in a blurrier generation of the objects in the image. In the subsequent fusion process, the low-contrast heat maps of noisy images are effective in suppressing the noisy images, while the heat maps of images containing co-salient objects are useful in overcoming overconfidence.

Figure A.7 further illustrates the effectiveness of stochastic features. Without the stochastic features $\mathcal{V}_i$, $\mathcal{F}_i$ might erroneously emphasize non-co-salient objects. Without proper guidance, the model may fall into the trap of excessive overconfidence. When stochastic features are integrated, the model effectively corrects the overconfidence in incorrect targets and suppresses the focus on non-co-salient regions.

Appendix A.3. Additional Implementation Details

We implement our model under the PyTorch framework with a single GeForce 3090 GPU. The transformer backbone is T2T-ViT$_t$-14 [4], which has lower computational complexity than VGG-16 [5] commonly used in CoSOD.

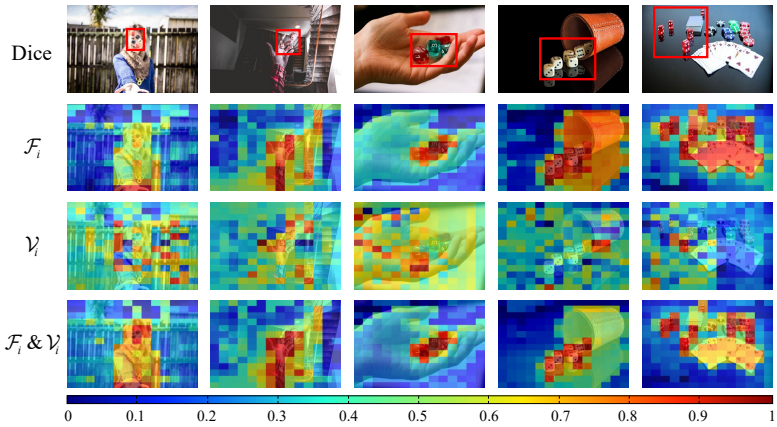


Figure A.7: Visualization of the effectiveness of stochastic features. The top row consists of images selected from the "Dice" group. The second row shows visualizations of the features $\mathcal{F}_i$ obtained from the CoSOD-TB. The third row displays the visualizations of the stochastic features $\mathcal{V}_i$ generated by the LVGB. The fourth row illustrates the visualizations of the features after integration during the transformer decoding phase.

We adopt a three-stage training scheme. In the first stage, COCO-SEG [6] is used to train the VQ-VAE and obtain the codebook. In the second stage, the generator is trained with frozen VQ-VAE weights. In the third stage, COCO-SEG [6] and DUTS [7] are combined to train the entire network. These two datasets contain 208,250 images from 369 categories with corresponding ground truths. During training, we randomly select two image groups and apply GSEM with $k = 1$ to exchange noisy images between them. Each training group contains $N = 5$ images, resized to $224 \times 224 \times 3$. For VQ-VAE training, we use 4-fold downsampling for 100 epochs with codebook size 128×384 (*i.e.*, 128 discrete vectors of 384 channels). The generator follows PixelCNN [8] and is trained for 50 epochs. The overall training lasts 60,000 steps with the Adam optimizer [9]. The hyperparameter in Eq. (7) is set to $\mu = 0.5$, and the commitment weight in Eq. (13) is set to $\lambda_0 = 0.25$. The loss weights in Eq. (19) are set to $\{\lambda_1 = 1, \lambda_2 = 0, \lambda_3 = 0\}$, $\{\lambda_1 = 0, \lambda_2 = 1, \lambda_3 = 0\}$, and $\{\lambda_1 = 0, \lambda_2 = 0, \lambda_3 = 1\}$ at the three training stages, respectively. Although training uses a two-group configuration with fixed $N = 5$ for GSEM and GPU-memory considerations, inference is not restricted by these settings. Because CoSOD-TB and the decoder are transformer-based and natively handle variable-length sequences, CoGSEM supports arbitrary group sizes (*e.g.*, $N = 2, 10, 20+$) without architectural

modification or retraining, and processes one image group at a time in unconstrained CoSOD scenarios.

Appendix A.4. Detailed Component Ablation

We provide a more detailed discussion of the component ablations corresponding to Table 3 in the main paper. Taking the results on CoCA as an example, removing GSEM leads to a noticeable decline in all four metrics. Specifically, $E_\phi^{\max}$ and $F_\beta^{\max}$ decrease by 1.3% and 1.8%, respectively, because GSEM effectively enhances the model's robustness through noise-introduced training. We also compare GSEM with its conference-version counterpart GEM: when using only GSEM or only GEM, their differences in the four metrics are 0.3%, 0.7%, 0.4%, and 0.7%, respectively, indicating that GSEM is more effective. Without LVGB, all four metrics are negatively affected. MAE increases from 0.092 to 0.094, S_α decreases from 0.740 to 0.731, $E_\phi^{\max}$ decreases from 0.820 to 0.812, and $F_\beta^{\max}$ drops from 0.621 to 0.614. This indicates that LVGB helps the model consider more possibilities from the input images, avoiding early-stage overconfidence and improving localization and segmentation of co-salient objects. Compared with its conference-version counterpart LVGB_p, LVGB better extracts stochastic components and outperforms LVGB_p by 0.1%, 0.5%, 0.5%, and 0.5% on the four metrics, respectively. CoSOD-TB is also crucial, as it models long-range dependencies in a sequence-to-sequence manner and enables precise segmentation of object details. When applied together with GSEM and LVGB, it further improves the four metrics by 0.6%, 1.9%, 1.9%, and 2.7%, respectively. It should be noted that applying any individual design in isolation may not always yield favorable results. For instance, when using GSEM alone, the MAE metric may deteriorate, because the introduced noise can disrupt shared group features and impair fine-grained detail capture if not compensated by LVGB and CoSOD-TB.

Appendix A.5. Sensitivity and Complexity Analysis

In this section, we conduct an ablation study on the hyperparameter μ in Eq. (7) to validate the effectiveness of the mixed difficulty metric, perform a sensitivity analysis on the codebook size K, and compare model complexity.

Table A.1 isolates the effects of the BDC metric and the binary metric. BDC excels at capturing high-dimensional semantic difficulty, while the binary metric handles low-level pixel difficulty. The hyperparameter μ balances the contributions of these two distinct difficulty spaces, where $\mu = 0$ corresponds to using the BDC metric alone. We provide more results to demonstrate the

Table A.1: Ablation studies on different μ settings.

Values of μ	CoCA				OWCoCA			
	MAE ↓	S_α ↑	$E_\phi^{\max}$ ↑	$F_\beta^{\max}$ ↑	MAE ↓	S_α ↑	$E_\phi^{\max}$ ↑	$F_\beta^{\max}$ ↑
0.0	0.111	0.728	0.813	0.611	0.121	0.729	0.667	0.439
0.1	0.105	0.732	0.809	0.612	0.117	0.732	0.680	0.448
0.3	0.105	0.731	0.810	0.609	0.101	0.751	0.681	0.453
0.5	0.092	0.740	0.820	0.621	0.090	0.754	0.686	0.469
0.7	0.093	0.730	0.817	0.621	0.090	0.754	0.690	0.460
0.9	0.098	0.734	0.811	0.617	0.099	0.742	0.683	0.462

Table A.2: Ablation studies on different codebook size K settings.

Values of K	CoCA				OWCoCA			
	MAE ↓	S_α ↑	$E_\phi^{\max}$ ↑	$F_\beta^{\max}$ ↑	MAE ↓	S_α ↑	$E_\phi^{\max}$ ↑	$F_\beta^{\max}$ ↑
64	0.097	0.733	0.813	0.616	0.093	0.747	0.684	0.460
128	0.092	0.740	0.820	0.621	0.090	0.754	0.686	0.469
256	0.092	0.739	0.819	0.620	0.088	0.751	0.688	0.465
512	0.091	0.741	0.822	0.620	0.089	0.753	0.685	0.468

Table A.3: Complexity analyses. For a fair comparison, the number of input images is set to 5 for all methods.

Methods	FLOPs (G)↓	Params (M)↓	Speed (fps)↑	CoCA $F_\beta^{\max}$ ↑
GICD(ECCV20)	364.7	278.0	40.8	0.504
GCoNet(CVPR21)	259.9	142.0	116.2	0.524
CADC(ICCV21)	330.0	392.8	18.0	0.548
DCFM(CVPR22)	251.9	142.3	84.4	0.598
CoGEM(CVPR23)	211.8	52.3	31.0	0.599
UniTR(TMM24)	347.4	167.6	30.3	0.567
TRNet(TCSVT25)	323.1	157.9	29.3	0.622
CoGSEM	213.2	54.1	30.0	0.621

performance sensitivity across $\mu \in \{0.0, 0.1, 0.3, 0.5, 0.7, 0.9\}$ in Table A.1, showing that $\mu = 0.5$ yields the optimal balance.

Table A.2 lists the model performance under different K values. With our default codebook size of 128, the codebook utilization rate is 85.2% with 19 dead codes. When K is set to 64, the utilization rate is 89.1% with 7 dead codes. When K is increased to 256 and 512, the utilization rates drop to 52.7% and 29.9%, with 121 and 359 dead codes, respectively, while no significant performance improvement is observed in Table A.2.

As shown in Table A.3, we compare FLOPs, parameter count, and inference time of CoGSEM against recent state-of-the-art models. While maintaining competitive performance, CoGSEM runs at a real-time speed of 30 fps with low FLOPs and few parameters, facilitating practical deployment.

Appendix A.6. Failure Cases

To provide a more comprehensive analysis of our model's performance, we also present some failure cases. From Figure A.8, it can be observed that our

model struggles with extremely challenging samples, including small objects, ghosting, and similar-looking distractors. When noisy images contain objects resembling the co-salient ones, the model also produces some false detections.

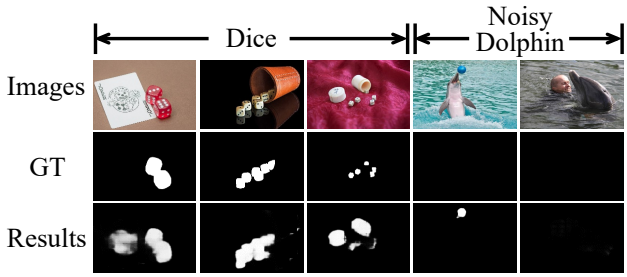


Figure A.8: Failure cases of our method sampled from OWCoCA.

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